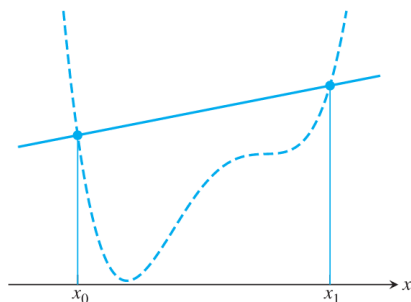


Integration

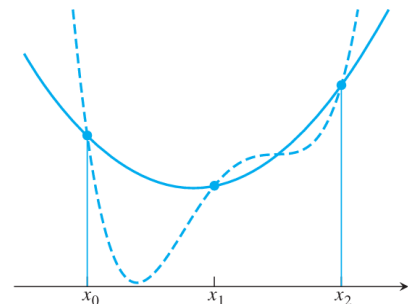
Recap

Trapezoid Rule



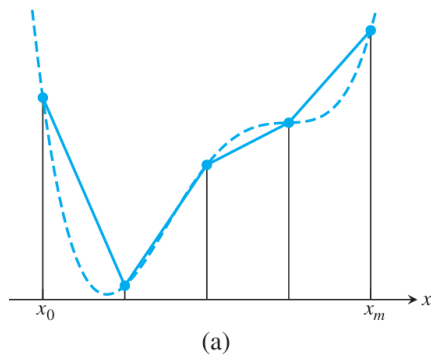
$$\int_{x_0}^{x_1} f(x) dx = \frac{h}{2}(y_0 + y_1)$$

Simpson's Rule



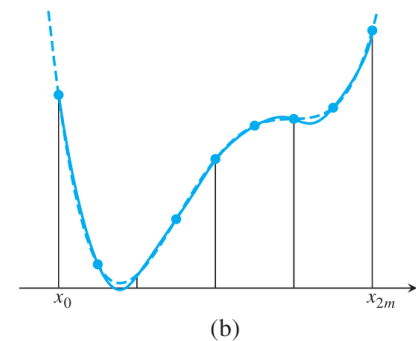
$$\int_{x_0}^{x_2} f(x) dx = \frac{h}{3}(y_0 + 4y_1 + y_2)$$

Composite trapezoidal



$$\int_a^b f(x) dx = \frac{h}{2} \left[f(a) + f(b) + 2 \sum_{i=1}^{m-1} f(x_i) \right]$$

Composite Simpson



$$\int_a^b f(x) dx = \frac{h}{3} \left[f(a) + f(b) + 4 \sum_{i=1}^m f(x_{2i-1}) + 2 \sum_{i=1}^{m-1} f(x_{2i}) \right]$$

Taylor remainder theorem:

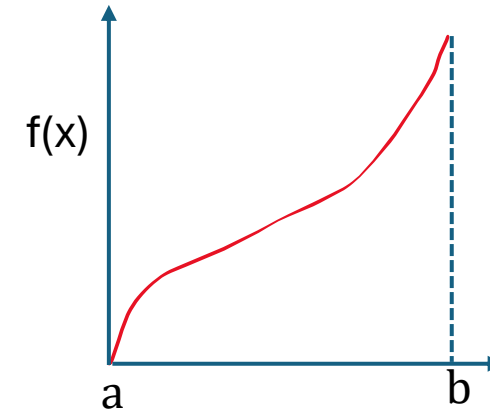
$$f(x) = f(m) + (x - m)f'(m) + R(x)[\text{Remainder}]$$

where, $R(x) = \frac{(x - m)^2}{2} f''(c)$ and c is a constant between x and m

The exact integral

$$\begin{aligned} I &= \int_{m-h/2}^{m+h/2} f(x) dx = \int_{m-h/2}^{m+h/2} \left(f(m) + (x - m)f'(m) + \frac{(x-m)^2}{2} f''(c) \right) dx \\ &= hf(m) + \frac{h^3}{24} f''(c) \end{aligned}$$

Mean Value Theorem



$$\int_a^b f(x)g(x)dx = f(c) \int_a^b g(x)dx$$

where, c is a constant between ' a ' and ' b '

$$f(c) = \int_a^b f(x)P(x)dx, \text{ where, } P(x) = \frac{g(x)}{\int_a^b g(x)dx}$$

(i). Numerically evaluate the integrals below using Trapezoidal and Simpson's methods.

$$(a) \int_0^1 x^2 dx \quad (b) \int_0^1 xe^x dx$$

Evaluate the exact integral and calculate the errors. Compute the maximum errors in each method using the error formulae and compare them with the earlier values.

(ii). Numerically evaluate the integral (b) using composite Trapezoidal and Simpson's methods with 5 panels.

$$|\text{Error in trapezoidal method}| = \frac{h^3}{12} f''(c)$$

$$|\text{Error in Simpson method}| = \frac{h^5}{90} f''''(c)$$

c is a constant between the integral limits $a = 0$ and $b = 1$